

UCab — Full-Stack MERN Cab Booking Platform

Solo Developer / Team Member: Shaik Sumiya Zainab | Project ID: N/A (Solo Track)

Phase 8: Project Demonstration — Step-by-Step Feature Walkthrough

Project Name: Cab Booking (`UCab`)

Project ID: `N/A (Solo Track Submission)`

Team Member / Solo Developer: Shaik Sumiya Zainab

1. Step-by-Step Live Demonstration Script

Follow this precise execution flow during evaluation to demonstrate all 22 functional features within a concise 3-minute window:

```
sequenceDiagram
    autonumber
    actor Evaluator as Evaluator / Grader
    actor SoloDev as Shaik Sumiya Zainab
    participant Client as React Frontend (Port 5173)
    participant Server as Express Backend (Port 8000)
    participant Store as Dual-Engine Store (data.json)

    SoloDev->>Server: npm start (Console shows Zero-Config Activated)
    SoloDev->>Client: npm run dev (Opens http://localhost:5173)
    SoloDev->>Client: Click "& One-Click Demo Rider Login"
    Client->>Server: POST /api/users/login
    Server->>Store: Lookup User (`sumiya@ucab.com`)
    Store-->>Server: User Profile + JWT
    Server-->>Client: Issue Bearer Token & Show Welcome Toast
    SoloDev->>Client: Navigate to /cabs, Filter Sedan, Click Book
    SoloDev->>Client: Enter Pickup & Drop, Apply Code `UCAB20`
    SoloDev->>Client: Check Refreshments (`+ 150`) & Charity (`+ 120`)
    SoloDev->>Client: Click "Confirm & Book Cab"
    Client->>Server: POST /api/bookings with Bearer Token
    Server->>Store: Save New Booking Trip
    Store-->>Server: Return Trip Confirmation
    Server-->>Client: Redirect to /home-user (Live GPS Progress Bar)
    SoloDev->>Evaluator: Point out Driver Assigned (`Vikram Sharma`)
    SoloDev->>Client: Open New Tab -> /admin/login -> Click Super Admin Demo
    SoloDev->>Client: Go to /admin/bookings -> Update Status (`Accepted` -> `Started`)
    Client->>Server: PUT /api/bookings/:id
    Server->>Store: Update Trip Status
    SoloDev->>Client: Switch to Rider Tab (`/bookings`) -> Status shows `Started`
    SoloDev->>Client: Click "Invoice" button on Completed Trip
    Client->>Evaluator: Render Thermal PDF Receipt Modal with Verification QR Code
```

2. Feature Checklist to Show Evaluator

- [x] Zero-Config Console Log: Show `Seeded/loaded 6 cars from automatic JSON persistence store`.
- [x] 1-Click Demo Login: Show instant session generation without typing passwords.
- [x] Promo Code Engine: Show `UCAB20` live 20% deduction on fare.
- [x] Perks Checkboxes: Show how `+ 150` (Refreshments) and `+ 120` (Donation) add to the total dynamically.
- [x] Live GPS Dispatch Tracker: Show the 4-stage tracking indicator and driver card.
- [x] Admin Dispatch & CRUD: Show how changing status in `/admin/bookings` syncs directly to the user's trip history.
- [x] QR Code Corporate Invoice: Show the printable invoice modal (`ReceiptModal.jsx`).